CASE STUDY

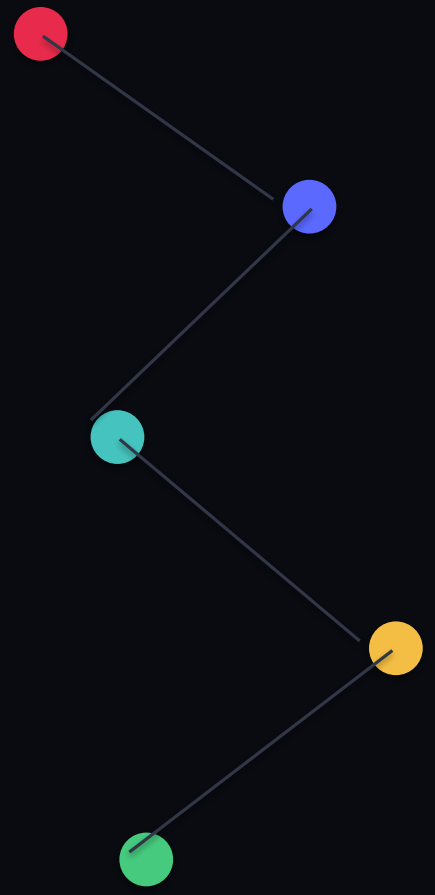
Zomato Demand Forecasting & Rider Planning

Turning historical order data into an operational decision.

PREDICTION → ACTION

Yaswanth Yadala

Case Study Presentation



Forecast • Localize • Plan • Explain

What Is This Case Study About?

The case study addresses a delivery-operations planning problem.

Food-delivery demand changes across days and locations. The operational challenge is not only knowing how much demand is coming — it is preparing the right delivery capacity where that demand is expected.

Core idea

This is therefore a decision-support case study — not just a forecasting exercise.

THE CASE STUDY CONNECTS

Historical orders	-	Understand demand patterns
Forecast	-	Estimate upcoming city demand
Locality allocation	-	Find where demand concentrates
Rider planning	-	Convert demand into capacity
Decision support	-	Turn results into an action

Problem Statement, Objectives & Business Value

PROBLEM STATEMENT

How can historical Zomato order data be used to forecast upcoming city demand and translate that forecast into locality-level rider requirements, so that delivery capacity can be planned more effectively?

OBJECTIVES OF THE CASE STUDY

- Forecast upcoming city-level order demand from historical patterns
- Identify where expected demand is concentrated across localities
- Allocate city demand into locality-level operational estimates
- Convert expected locality demand into active and standby rider recommendations
- Assess priority, operational risk and decision confidence
- Present the complete analysis through an interactive decision-support interface

BENEFITS OF SOLVING THE PROBLEM

- Better alignment between expected demand and rider capacity
- Reduced risk of under- or over-deployment
- Improved locality-level operational planning
- Clearer prioritization of demand hotspots
- More interpretable decisions through risk and confidence indicators
- A reusable decision-support workflow rather than a forecast alone

Central idea: move from “How much demand is expected?” to “What operational action should be taken?”

The Problem We Are Trying to Solve

A city-level forecast answers “how much?” — but operations also needs to know “where?” and “what next?”

IF WE STOP AT FORECASTING

53

Expected city orders

Still unknown:
Where? Riders? Standby? Risk?

WHAT OPERATIONS ACTUALLY NEEDS

- Where is demand concentrated?
- How much demand is expected in each locality?
- How many riders should be deployed?
- How much standby capacity is needed?
- How reliable is the recommendation?

Problem statement: How can historical order data be used to forecast upcoming demand and translate it into locality-level rider requirements?

Why Is Solving This Problem Necessary?

Delivery capacity has to match demand — but both under-planning and over-planning have consequences.

TOO FEW RIDERS

High demand + insufficient capacity

- Potential delivery delays
- Higher rider workload
- Poor operational response

TOO MANY RIDERS

Low demand + excess capacity

- Inefficient resource utilization
- Unused delivery capacity
- Higher planning inefficiency

So the need is not simply to predict demand — it is to make the prediction useful for capacity planning.

The Questions That Drive the Case Study

CITY

How many orders are expected tomorrow?
How does that compare with history?
Is demand LOW / MEDIUM / HIGH?

LOCALITY

Where is demand concentrated?
How much demand is expected locally?
Is there enough historical data?

RIDER

How many active riders are needed?
How many standby riders?
Which areas need more availability?

DECISION

What is the operational risk?
How confident is the recommendation?
Why did the system recommend it?

What I Proposed to Solve It

Instead of building an isolated forecast model, I proposed an end-to-end demand-to-decision workflow.



The key transition

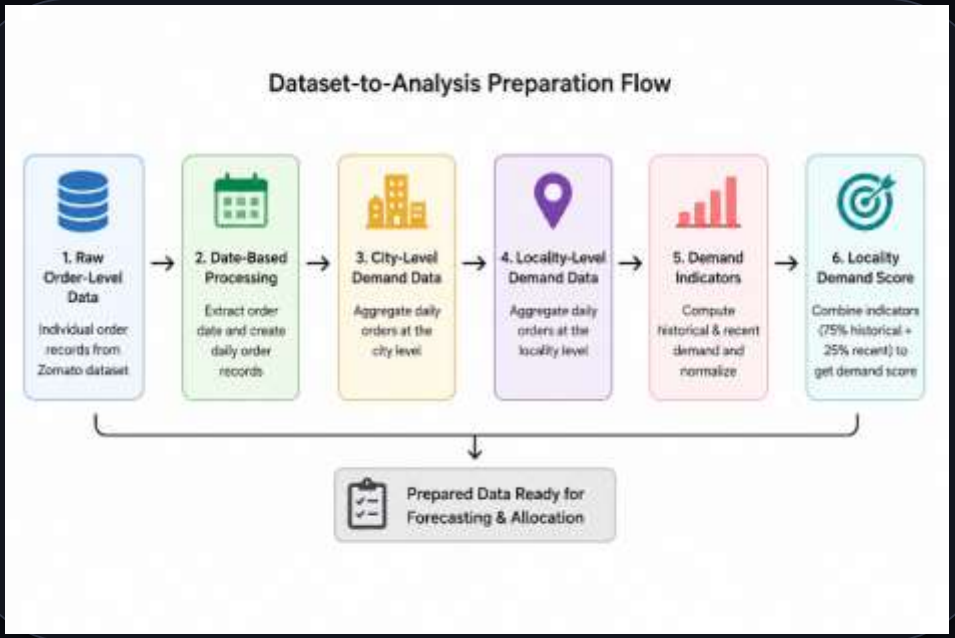
"How much demand should I expect?"

↓

The final question

"What should I do with that demand?"

How the Data Was Turned into Useful Signals



Two demand views were needed

- Historical demand → stable long-term pattern
- Recent demand → captures recent behaviour
- Both were normalized before combining

LOCALITY DEMAND SCORE

75% Historical + 25% Recent

Data sufficiency became part of the solution

At least 30 original historical records are required for locality-level forecasting.

< 30 records → Forecast unavailable

Step 1 — Forecast the Upcoming City Demand

The first analytical question is: how much demand is expected for the upcoming operational day?

1. **Historical demand**
   baseline
2. **Recent behaviour**
   7-day view
3. **Forecast**
   next-day demand
4. **Classification**
   LOW / MEDIUM / HIGH



Example from testing

53	121.64	LOW
Expected orders	Historical avg	Demand status

53 < 121.64 → forecast classified as LOW.

Step 2 — Find Where the Demand Is Concentrated

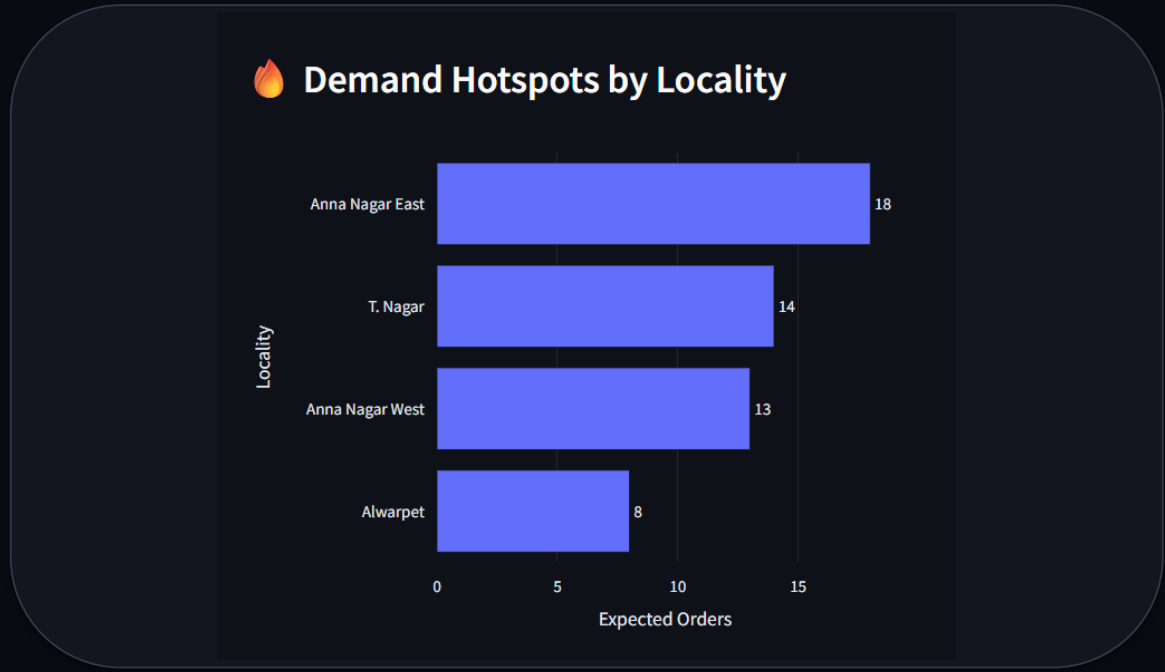
A city total is not enough because riders operate in specific areas.

LOCALITY ALLOCATION LOGIC

Rank localities by demand score

↓

Allocate forecast to priority zones



Planning assumptions

- Top 10 localities used as active zones
- Approximately 90% of city forecast allocated to these zones
- Rounding correction keeps allocation consistent

Output: demand concentration + locality-level expected orders

Step 3 — Convert Locality Demand into Rider Capacity

This is where the case study moves from demand estimation to an operational recommendation.



EXAMPLE

13 expected orders

→ 4 active + 2 standby

Expected orders

locality demand

Active riders

normal deployment

Standby riders

buffer for fluctuations

Recommendation

human-readable action

Important: rider counts are planning recommendations based on the case-study rules, not exact real-world staffing requirements.

What Did the Case Study Find?

The locality-level analysis exposed where operational attention would be most useful.



Anna Nagar East	**18 expected orders**	Highest hotspot in the shown test output
T. Nagar	**14 expected orders**	Second-highest shown locality
Anna Nagar West	**13 expected orders**	High-priority demand concentration
Alwarpet	**8 expected orders**	Lower than the top hotspots in the shown output

Shown values are from the report's testing visualization.

The Final Analytical Layer — From Numbers to Decisions

The outputs are combined so an operations user does not have to interpret each calculation separately.

RISK

LOW / MEDIUM / HIGH

A simple planning signal based on demand and operational conditions.

PRIORITY

Where to focus

Highlights localities with greater expected demand.

CONFIDENCE

75% example

Communicates how strongly the available evidence supports the recommendation.

Why this layer matters

- Explains what requires attention
- Separates confidence from the raw forecast
- Produces a recommendation rather than a number alone

END RESULT

Forecast → Plan → Explain → Act

Why Was an Interface Needed?

The interface was proposed as the delivery layer of the case study — not as a separate project.

WITHOUT A DECISION INTERFACE

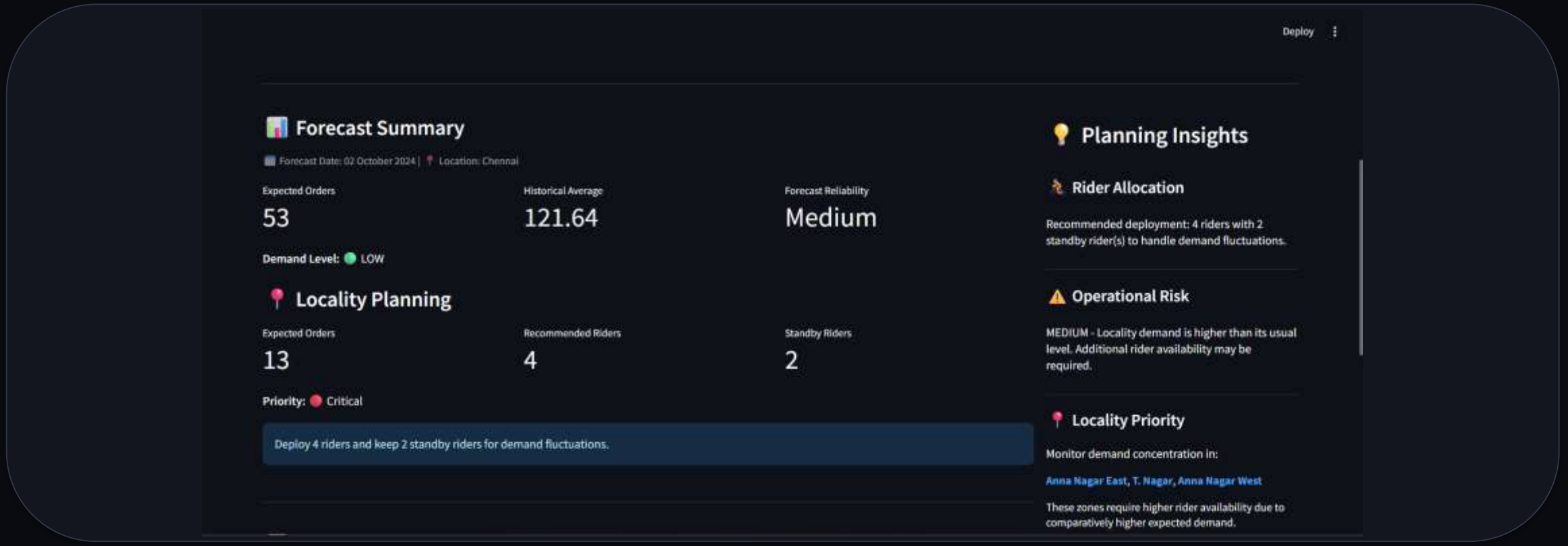
- User must inspect forecast values separately
- Locality demand is disconnected from rider planning
- Risk and confidence are harder to interpret
- Final action has to be manually inferred

WITH THE PROPOSED INTERFACE

- Select city + locality
- Generate forecast
- See locality planning + visual demand
- Review risk, priority and confidence
- Read the operational recommendation

Purpose: make the analytical workflow usable, interpretable and actionable in one place.

How the Interface Supports the Case Study



1	**Location**	Choose city + locality
2	**Forecast**	See expected demand + status
3	**Planning**	Riders + standby + priority
4	**Insights**	Trend + hotspots + risk + confidence

The interface mirrors the analytical logic of the case study.

A Complete Example: From Forecast to Action

53	121.64	13	4 + 2	75%
City forecast	Historical average	Locality forecast	Riders + standby	Decision confidence

WHAT THE USER GETS

Deploy 4 riders and keep 2 standby riders for demand fluctuations.

Supported by locality demand, priority/risk information and the decision-confidence layer.

PREDICTION → OPERATIONAL ACTION

The report's example illustrates the intended end-to-end decision flow.

What I Learned by Testing the Case Study

Several technically valid outputs were not operationally useful — so the logic was refined.

City forecast → locality plan	A city total alone was insufficient for rider planning.
Few records → unavailable	Insufficient data must not look like zero demand.
Fixed rider rule → demand-linked	Rider recommendations were refined to relate more directly to expected demand.
City risk → locality risk	Risk logic was corrected to reflect the selected locality.
Missing match → safe status	Unsupported locality recommendations now return an unavailable status.

Key lesson: a useful analytical system must consider data quality, assumptions, interpretability and the decision that follows.

What Can We Conclude from the Case Study?

The final outcome is more than a predicted order count.

Forecast

Estimate upcoming city demand

Localize

Identify demand concentration

Plan

Recommend active + standby riders

Assess

Priority, risk and confidence

Recommend

Communicate an operational action

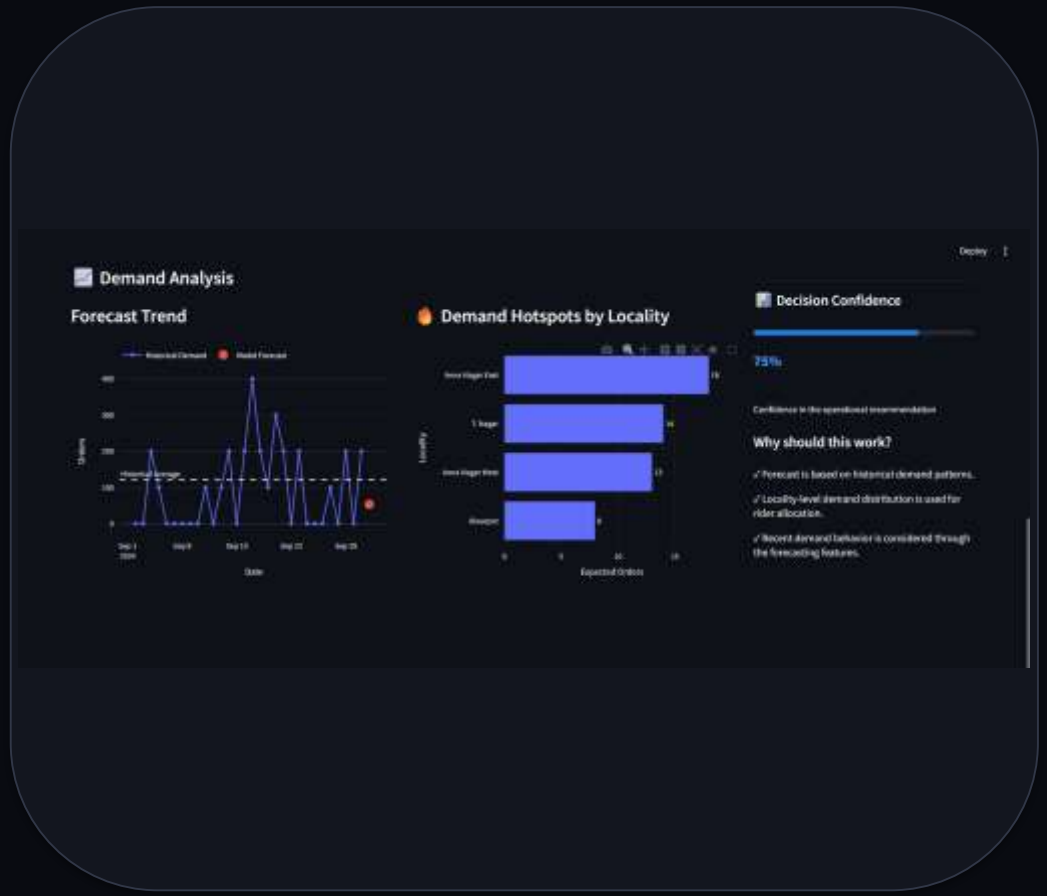
The case study successfully connects historical demand forecasting with locality-level resource planning and communicates the resulting recommendation through an interactive decision-support interface.

Scope reminder: rider counts are recommendations based on the assumptions and rules defined in the case study.

THANK YOU

From Demand Forecast to Action

Zomato Demand Forecasting & Rider Planning



Yaswanth Yadala